

Synthetic Catastrophe Model Overview

This document is a synthetic, illustrative overview of a catastrophe risk model created for demonstration purposes only. It does not describe any real vendor model or real insurance product.

The model estimates potential losses from natural catastrophe events such as hurricanes, earthquakes, and severe convective storms by combining hazard, exposure, and vulnerability modules.

Exposure characteristics used by the model include building construction type, occupancy class, year built, replacement cost value, and geographic location at the address level.

Hazard Module Assumptions

The hazard module simulates thousands of stochastic event years to represent the full range of possible catastrophe activity.

For hurricane risk, the model assumes wind speed decay based on distance from landfall, terrain roughness, and storm size.

For earthquake risk, the model assumes ground motion attenuation based on fault distance, soil type, and moment magnitude.

Vulnerability and Loss Estimation

Vulnerability functions translate hazard intensity, such as peak wind gust or peak ground acceleration, into expected damage ratios for a given construction type.

The model applies these damage ratios to the insured replacement cost value to estimate gross loss before policy terms are applied.

Policy conditions such as deductibles, limits, and coinsurance are then applied to estimate the net loss to the insurer.

Exposure Guidelines

Exposure data quality directly affects model accuracy. Address-level geocoding is strongly preferred over ZIP-code-level geocoding.

Construction type should be verified against underwriting documentation rather than assumed from occupancy class alone.

Replacement cost values should be updated at each renewal to reflect current construction costs and avoid systematic underinsurance.

Model Validation Notes

This synthetic model has not been validated against real historical loss experience and should not be used for any actual underwriting, pricing, or reserving decision.

In a real production setting, model validation would compare modeled average annual loss and exceedance probability curves against actual historical loss experience where available.

Known limitations of this illustrative model include the absence of storm surge, demand surge, and post-event loss amplification.